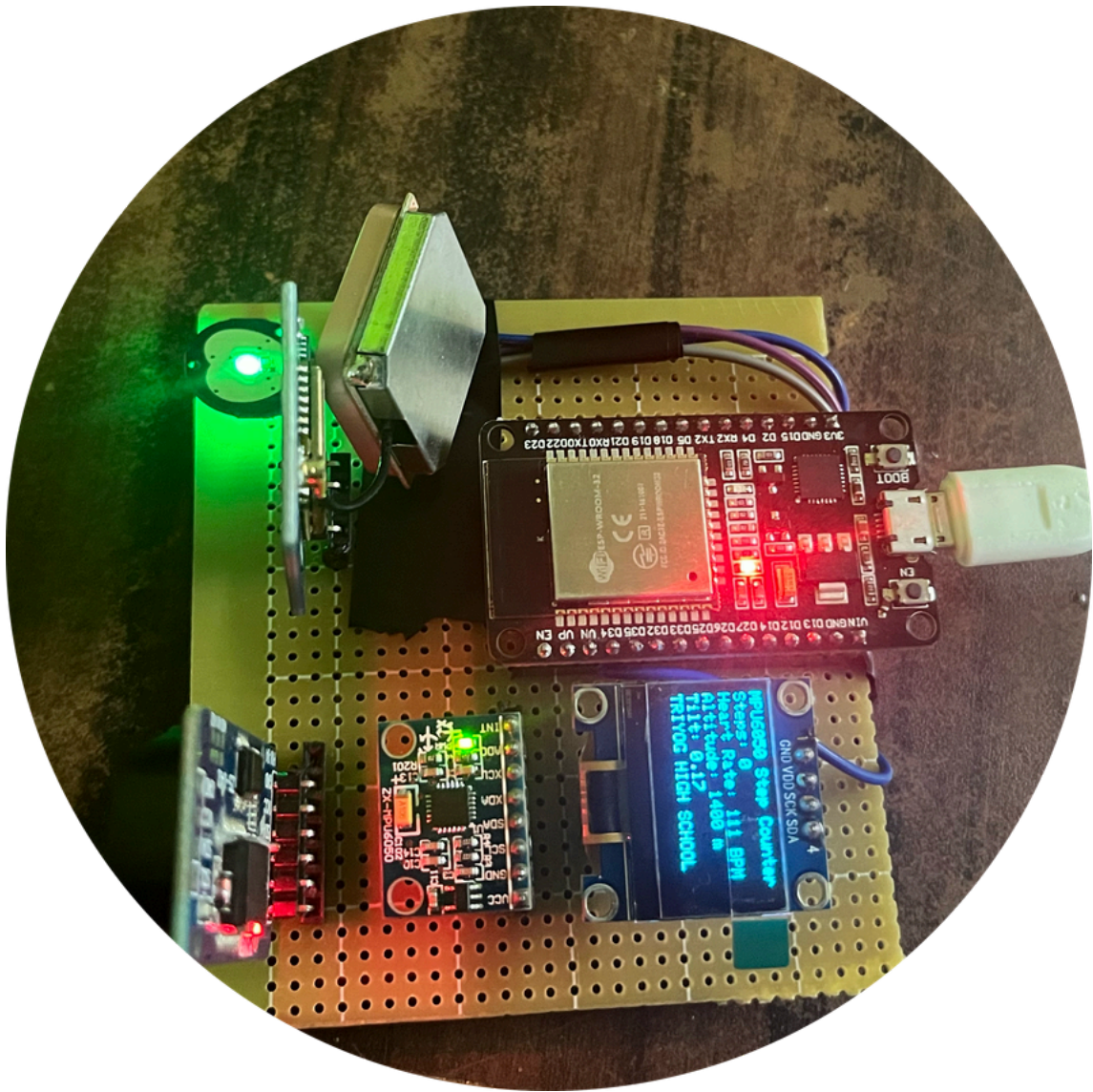


Standalone Health & Activity Monitor with ESP32



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GitHub : <https://github.com/sourya3/Swastha-Lab.git>

Key Project Features

Cardiovascular Monitoring: Real-time Heart Rate (BPM) tracking using optical photoplethysmography (PPG).

Pedometer & Motion Analysis: 3-axis acceleration vector synthesis for step detection and movement logging.

Independent Timekeeping: Sub-second RTC logging without requiring Wi-Fi connection or NTP synchronization.

Geographic Elevation Mapping: Real-time altitude above sea level and coordinate tracking via satellite GPS streams.

Low-Power Display UI: Compact graphical interface delivering live data updates across a monochrome OLED display.

Component	Function / Role	Operating Protocol
ESP32 Dev Board	Central MCU / Dual-core processing	Core Controller
0.96" SSD1306 OLED	Real-time graphical user interface	I ² C (0x3C)
MPU6050 IMU	6-axis motion and step detection	I ² C (0x68)
DS3231 RTC Module	Precise date and timestamp generation	I ² C (0x68 / 0x57)
NEO-6M GPS Module	Geographic coordinate & altitude parsing	Hardware Serial (UART2)
TZT Pulse Sensor	Optical finger-touch heart rate sampling	Analog ADC (GPIO 34)

Hardware Module Trade-Off Analysis

1. ESP32 Microcontroller

- **Advantages:** Dual-core processor (240 MHz) provides fast multi-tasking for concurrent sensor processing, built-in Wi-Fi/BLE for optional wireless expansion, and flexible GPIO routing.
- **Disadvantages:** Higher active current draw (~80–160 mA) compared to simple 8-bit microcontrollers, and ADC2 pins cannot be used while Wi-Fi is active (hence using GPIO 34 on ADC1).

2. TZT Pulse Sensor (Analog PPG)

- **Advantages:** Ultra-low cost, lightweight, and simple analog voltage output requiring no complex communication protocols.
- **Disadvantages:** Highly vulnerable to motion artifacts and ambient light noise; requires algorithm-level filtering and baseline calibration to avoid false pulse counts.

3. MPU6050 6-Axis IMU

- **Advantages:** High-precision motion detection with low power consumption; combines 3-axis accelerometer and 3-axis gyroscope on a single I2C bus node (0x68).
- **Disadvantages:** Raw accelerometer signals pick up structural vibrations, requiring digital debouncing or moving-average filters for pedometer applications.

4. DS3231 Real-Time Clock (RTC)

- **Advantages:** Integrated Temperature-Compensated Crystal Oscillator (TCXO) maintains sub-second time accuracy (± 2 ppm drift) even in varying ambient temperatures.
- **Disadvantages:** Occupies I2C address 0x68 (sharing the default address with MPU6050, requiring ADO pin management if modifying addresses).

5. NEO-6M GPS Module

- **Advantages:** Autonomous location and altitude lock with onboard EEPROM to save setup parameters; standard UART output works with any serial interface.
- **Disadvantages:** Slow cold-start fix time (30–45 seconds) in indoor or heavily shaded environments; high peak current draw during satellite acquisition (~45 mA).

6. 0.96" SSD1306 OLED Display (128×64)

- **Advantages:** High contrast with self-illuminating pixels (no backlight required), crisp readability, and ultra-low power consumption when displaying sparse text.
- **Disadvantages:** Small physical footprint limits the amount of concurrently rendered metrics; prone to OLED pixel burn-in if static graphics remain unchanged for extended periods.

Safety Considerations :

1. Medical & Diagnostic Disclaimer

- **Non-Medical Grade Device:** This system is an experimental, open-hardware prototype designed solely for educational, research, and non-critical monitoring purposes.
- **Clinical Limitation:** The TZT pulse sensor and algorithms are not FDA-approved or clinically certified. Output values (BPM) must not be used for medical diagnosis, treatment decisions, patient monitoring in critical conditions, or emergency response systems.

2. Electrical & Power Safety

- **Battery Handling (Li-Ion/Li-Po):** If powered via external 3.7V Li-Ion batteries (such as 18650 cells), use an integrated protection circuit module (PCM/BMS, e.g., TP4056 with overcharge, over-discharge, and short-circuit protection) to prevent thermal runaway or fire hazards.
- **Voltage Regulation:** Ensure strict regulation to 3.3V logic levels for the ESP32 and attached peripherals. Supplying 5V directly to 3.3V-rated GPIO pins can permanently damage components or cause overheating.

System Connections

OLED Display

<u>OLED</u>	<u>ESP 32</u>
VCC	3.3V
GND	GND
SDA	GPIO 21
SCL	GPIO 22

MPU6050 IMU

<u>MPU 6050</u>	<u>ESP 32</u>
VCC	3.3V
GND	GND
SDA	GPIO 21
SCL	GPIO 22

DS3231 RTC Module

<u>DS3231</u>	<u>ESP 32</u>
VCC	3.3V
GND	GND
SDA	GPIO 21
SCL	GPIO 22

NEO-6M GPS Module

<u>GPS Pin</u>	<u>ESP 32</u>
VCC	3.3V
GND	GND
TX	GPIO 16
RX	GPIO 17

TZT Pulse Sensor

<u>Sensor Pin</u>	<u>ESP 32</u>
VCC	3.3V
GND	GND
Signal (S)	GPIO 34

Required Software Libraries & GitHub Links :

1. Adafruit Industries. Adafruit SSD1306 Library. https://github.com/adafruit/Adafruit_SSD1306
 2. Adafruit Industries. Adafruit GFX Library. <https://github.com/adafruit/Adafruit-GFX-Library>
 3. Adafruit Industries. Adafruit MPU6050 Library. https://github.com/adafruit/Adafruit_MPU6050
 4. Adafruit Industries. Adafruit Unified Sensor Library. https://github.com/adafruit/Adafruit_Sensor
 5. Adafruit Industries. RTCLib. <https://github.com/adafruit/RTCLib>
 6. Mikal Hart. TinyGPSPlus Library. <https://github.com/mikalhart/TinyGPSPlus>
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1. Arduino. Wire (I²C) Library Reference. <https://www.arduino.cc/reference/en/language/functions/communication/wire/>
 2. Espressif Systems. ESP32 HardwareSerial Reference. <https://docs.espressif.com/projects/arduino-esp32/en/latest/api/uart.html>